

Problem Solving

11 September 2026 09:39

ALGORITHM

In our day-to-day life, we perform activities by following certain sequence of steps. Examples of activities include. Getting ready for school, making breakfast, riding a bicycle, wearing a tie, solving a puzzle, and so on. To complete each activity, we follow a sequence of steps. Suppose following are the steps required for an activity 'Riding a bicycle'.

1. Remove the bicycle from the stand.
2. Sit on the seat of the bicycle.
3. Start peddling.
4. Use brakes whenever needed. And,
5. Stop on reaching the destination.

Let us now find greatest common divisor (GCD) of two numbers 45 and 54.

1. Find the numbers. (divisors) which can divide the given numbers:
Divisors of 45 are: 1, 3, 5, 9, 15 and 45
Divisors of 54 are: 1, 2, 3, 6, 9, 18, 27 and 54
2. Then find the largest common number from these two lists.
3. Therefore, GCD of 45 and 54 is 9.

Hence, it is clear that we need to follow a sequence of steps to accomplish the task. Such a finite sequence of steps required to get the desired output is called an algorithm. It will lead to the desired result in a finite amount of time if followed correctly. Algorithm has a definite beginning and the definite end, and consist of a finite number of steps.

Why do we need an Algorithm?

The programmer first prepares a road map of the program to be written before actually writing the code. Without a road map, the programmer may not be able to clearly visualize the instructions to be written and may end up developing a program which may not work as expected.

Such a road map is nothing but the algorithm which is the building block of a computer program. For example, searching using a search engine, sending a message, finding a word in a document, booking a taxi through an app, performing online banking, playing computer games are all based on algorithms.

Writing an algorithm is mostly considered as a first step to programming. Once we have an algorithm to solve a problem, we can write the computer program for giving instructions to the computer in a high level language. If the algorithm is correct, computer will run the program correctly every time. So the purpose of using an algorithm is to increase the reliability, accuracy and efficiency to obtain solutions.

Characteristics of a good algorithm:

- **Precision:** The steps are precisely stated or defined.
- **Uniqueness:** Results of each steps are uniquely defined and only

dependent on the input and the result of the preceding steps.

- **Finiteness:** The algorithm always stops after a finite number of steps.
- **Input:** The algorithm receives some input.
- **Output:** The algorithm produces some output.

While writing an algorithm, it is required to clearly identify the following:

- The input to be taken from the user.
- Processing or computation to be performed and to get desired result.
- The output desired by the user.

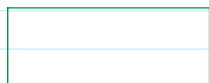
Flow chart- Visual representation of algorithms

A flow chart is a visual representation of an algorithm. A flow chart is a diagram made up of boxes, diamonds and other shapes connected by arrows. Each shape represents a step of the solution process, and the arrow represents the order or link among the steps.

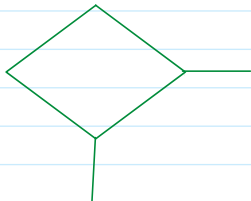
There are standardized symbols to draw flow chart.



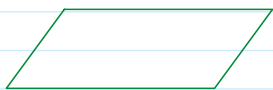
Start/End → Also called "Terminator" symbol. It indicates where the flow starts and ends.



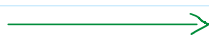
Process → Also called "Action symbol", it represents a process, action, or a single step.



Decision → A decision or branching point, usually a yes/no or true/false question is asked and based on the answer, the path gets split into two branches.



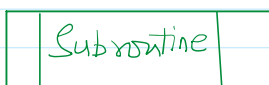
Input/Output → Also called data symbol, this parallelogram shape is used to input or output data.



Arrow → Connector to show order of flow between shapes.



Connector → When our flowchart is big then we use connector to join with another page.



Subroutine/Function → use to start a subroutine or function which is called in the main flow chart.

Write an algorithm to find the square of a number.

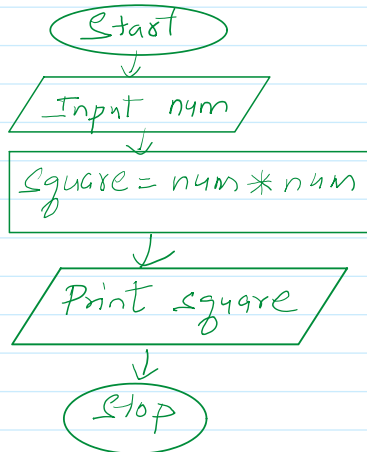
Before developing an algorithm, let us first identify the input, process and output.

- Input: Number whose square is required.
- Process: Multiply the number by itself to get its square.
- Output: Square of the number.

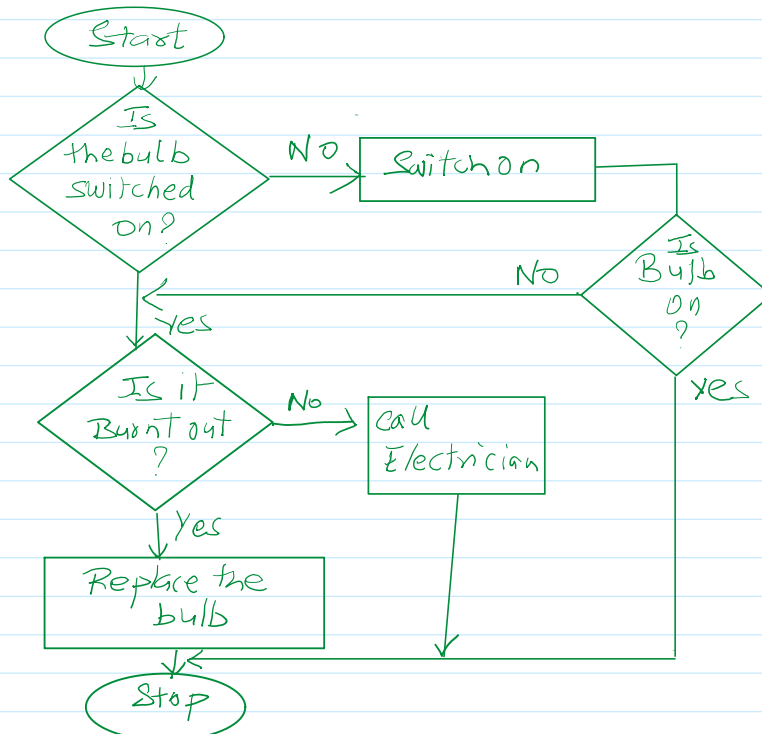
Algorithm to find the square of a number:

1. Input a number and store it to num.
2. Compute $\text{num} * \text{num}$ and store it in a square.
3. Print square.

The algorithm to find a square of a number can be represented pictorially using flow chart as shown in figure below:



Draw a flow chart to solve the problem of a non-functioning light bulb.



Pseudocode: A pseudo code (pronounced Soo-doh-kohd) Is another way of representing an algorithm. It is considered as a non-formal language

that helps programmer to write algorithms. It is a detailed description of instructions that a computer must follow in a particular order. It is intended for human reading and cannot be executed directly by the computer. No specific standard for writing a pseudocode exist. The word pseudo means not real, so pseudo code means not real code. Following are some of the frequently used keywords while writing pseudocode:

- INPUT
- COMPUTE
- PRINT
- INCREMENT
- DECREMENT
- IF/ELSE
- WHILE
- TRUE/FALSE

Write an algorithm to display the sum of two numbers entered by the user using both pseudocode and flowchart.

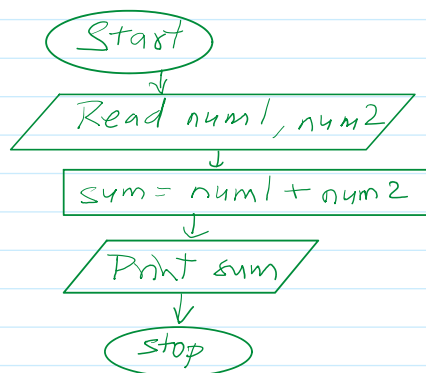
Pseudocode for the sum of two numbers will be:

INPUT num1.

INPUT num2.

COMPUTE $\text{sum} = \text{num1} + \text{num2}$

PRINT sum



Homework

1. Write an algorithm to calculate area and perimeter of a rectangle using both pseudo code and flow chart.
2. Write an algorithm to check whether a number is odd or even. Draw the flow chart also.
3. Write a pseudo code and draw a flow chart where multiple conditions are checked to categorize a person as either child (less than 13) Teenager (greater than equal to 13 but less than 20) or Adult (greater than equal to 20), Based on ages specified.

Input: Age

Process: Check age as per given criteria.

Output: Print either "Child", "Teenager", or "Adult".

